Exchange-Traded Funds (ETFs): The Future of Investing - Demand Side Dynamics

Swapnil Tungare – University of Delaware, Lerner College of Business & Economics
FINC 612 – Capital Markets & Financial Institutions

Introduction

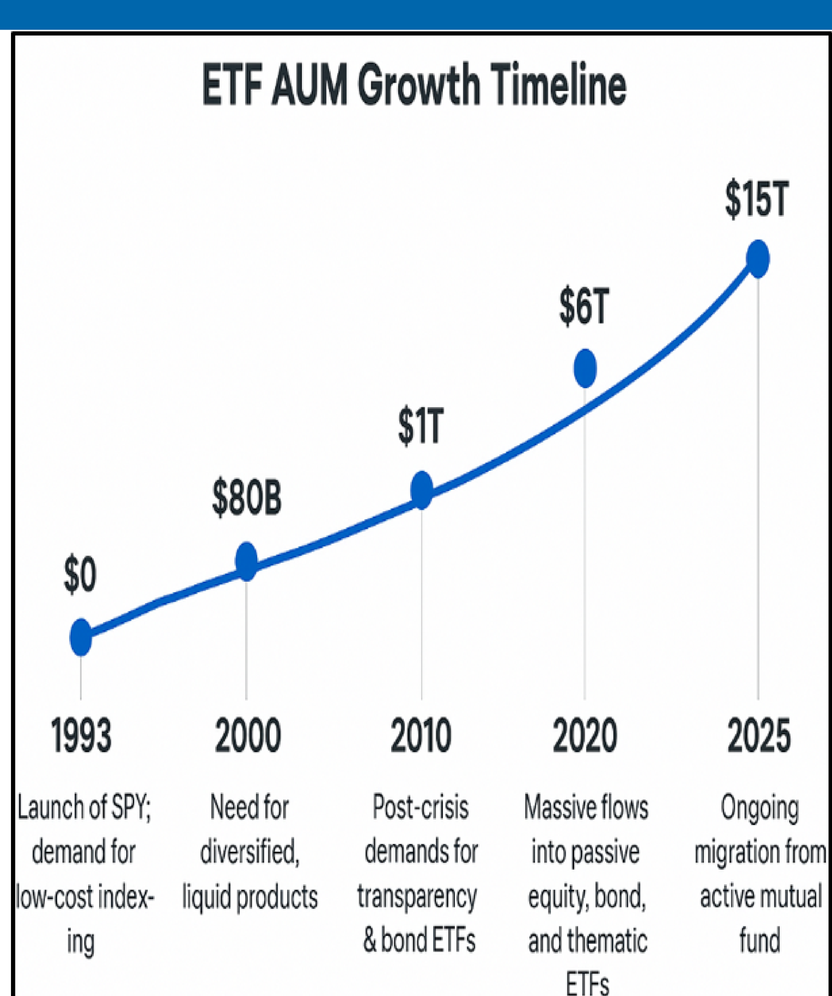
Exchange-Traded Funds (ETFs) have grown from $0 in 1993 to nearly $15T in global AUM by 2025, becoming the dominant investment vehicle for both retail and institutional investors. This poster examines why demand shifted dramatically from mutual funds to ETFs, focusing on real-time investor behaviour and structural product advantage*s*.

Key drivers of ETF demand:

- Intraday liquidity and continuous pricing.
- Lower fees and tighter spreads.
- High transparency and tax efficiency.
- Rapid adoption through technology and retail investing apps

Research Objective: To analyse the demand-side growth of ETFs using order flow evidence and ratio-based comparison with mutual funds.

ETF AUM Growth

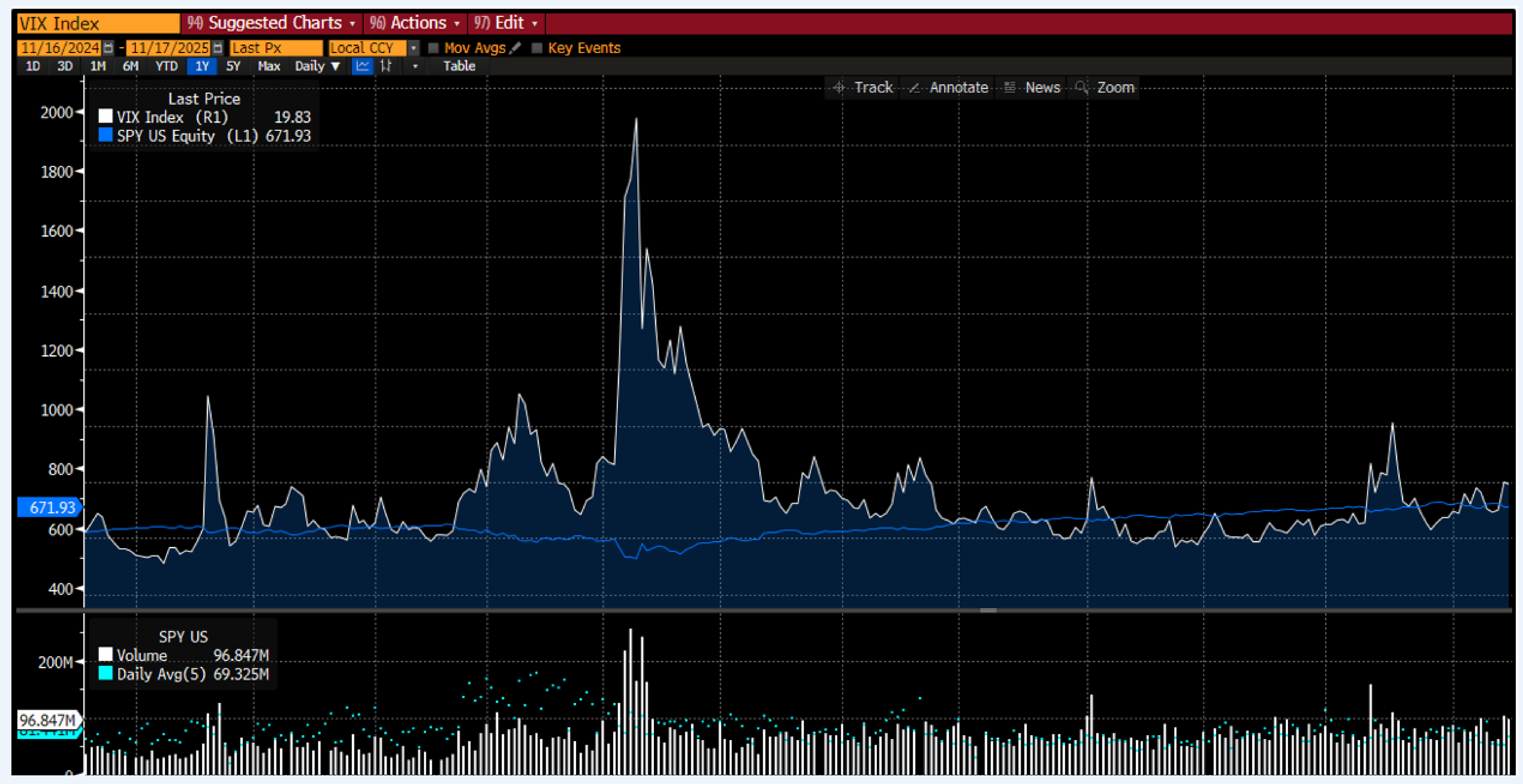


Methodology

A. Behavioural Analysis
Purpose: Identify real-time investor demand and trading behaviour.
(**Dataset**: Bloomberg :–FFLO- net inflow/outflow, daily volume, turnover, VIX correlation)

B. Mutual Fund vs ETF Ratio
Purpose: Explain long-term structural migration from mutual funds to ETFs.
(**Ratios Used**: Expense ratio, bid–ask spread, volume/AUM, tax cost ratio, transparency frequency.)

Behavioural Analysis



Source: TradingView. Chart shows SPY daily volume and VIX volatility overlay.
VIX ↑ → ETF demand ↑ Positive correlation **ETF hedging** + liquidity during volatility.
ETFs attract investors because they provide liquidity during market stress.

Bloomberg Net Flows: Real-Time Investor Demand" Bloomberg Net Flow data highlights a major demand-side shift toward ETFs. Equity ETFs (blue) drive the majority of new inflows, pushing cumulative flows above $6.6T by 2025. This confirms that ETFs—especially equity-based—are the preferred, liquidity-driven vehicle for new investor capital

ETFs vs. Mutual Funds: The Structural Case

Metrics	SPY (ETF)	VFIAX (MF)	Interpretation
Expense Ratio (Cost advantage)	0.09%	0.04%	Exception: ETF ER > MF ER
Bid–Ask Spread (Liquidity advantage)	$0.01–0.02	(NAV only)	ETF: intraday liquidity – tight spread
Turnover / Volume (Trading efficiency)	High (daily trading)	Medium	ETF: trading efficiency
Tax Cost Ratio (Structural efficiency)	0.5%	1.2%	ETF: lower tax drags than MF.
Transparency (Daily vs. quarterly disclosure)	Daily	Quarterly	ETF: more transparency increases investor trust and adoption.

ETF–Mutual Fund structural comparison highlights why ETFs maintain long-term cost and efficiency advantages. SPY has a slightly higher expense ratio than some mutual funds due to its UIT structure, but superior liquidity, tighter spreads, daily transparency, and in-kind creation/redemption make its total trading cost lower. Ratios such as spreads, turnover, transparency frequency, and tax cost demonstrate persistent structural benefits favoring ETFs over mutual funds.

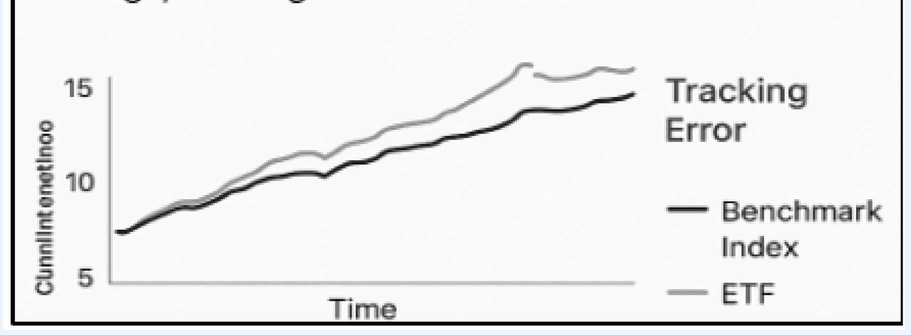
Transaction Costs

ETFs trade like stocks, so investors face bid-ask spreads, brokerage fees, and potential market impact. These costs can reduce net returns, especially in smaller or less liquid ETFs.

ETF Type	Example ETF	Ask Price	Spread (bps)
Large-cap ETF	SPY	$450.00	0.44
Small-cap ETF	IWM	$220.00	4.55

Tracking Error

Measures how closely an ETF follows its benchmark. Differences arise from management fees, sampling of index constituents, dividend timing, and cash holdings, causing the ETF to deviate from the index.



Backtesting

Structural Timing Advantage Test: SPY vs VFIAX **Backtest: 6-Month Investment of $10,000**

- **SPY** Price Buy: $586.02 → Shares: 17.07 Current Value: 17.07 × 671.93 = **$11,468**
- **VFIAX** Price Buy: $543.34 → Shares: 18.41 Current Value: 18.41 × 622.46 = **$11,455**

Why SPY earns 13$ more
SPY trades intraday, letting investors buy at real-time prices.**VFIAX is priced once per day** at NAV, causing small timing delays. Over short periods, this **structural timing advantage** leads to slight return differences.

ETF Investing

ETF Investing across countries
USA: Highest ETF adoption; strong inflows due to low fees, transparency, and wide product availability.
UK/Europe: Growing steadily; investors prefer ETFs for equity and bond exposure, though mutual funds still dominate.
Japan: ETF adoption slower; retail investors are cautious, but institutional flows are increasing.
Emerging Markets (Brazil, India): ETFs small but growing rapidly; adoption driven by cost efficiency and global index exposure.

ETF - Market sensitivity change in inflow and outflow due to;

- Market volatility ↑ Flows ↑ (Hedging)
- Interest rate change ↑ Demand ↓ (Yield shift)
- Macro-economic news Flows ↑ ↓(Uncertainty)
- Global risk ↑ Flows ↑(Safe sector) ↓(Risky Markets)
- Investors behaviour ↑ Flows ↑ (Short term)

Conclusion

ETF demand is driven by a combination of behavioral and structural forces. During volatility, investors favor ETFs for their liquidity, transparency, and tighter trading efficiency. Structurally, ETFs outperform mutual funds through lower costs, tax advantages, and superior execution. Together, these factors explain the long-term shift in global investor preference, positioning ETFs as the dominant investment vehicle of the future.